AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM298
Operating Systems

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 15/Apr./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 1.5 Hours

Instructions:

- Total Marks: **30**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are NOT allowed.

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Part 1:Multiple Choice Questions: [5 marks]

Question 1: Choose the most correct answer.

1. Which component of a computer system provides basic computing resources?
   a. Operating system
   b. Hardware
   c. Users
   d. Application Program
2. Which of the following best describes the kernel?
   a. Program always running.
   b. Middleware layer.
   c. User Application.
   d. Device controller.
3. In a vectored interrupt system, the interrupt vector contains:
   a. Device buffers.
   b. CPU registers.
   c. Polling algorithm.
   d. None of the choices.
4. Which OS service allows users to interact with the system?
   a. Resource allocation
   b. Error detection
   c. User interface
   d. Logging
5. The program counter stores:
   a. Process ID
   b. Register values
   c. Next instruction address
   d. Both b and c

Part 2: Short Essay Questions: [15 marks]

Question 2: [2 marks]

Why are threads referred to as lightweight processes? Discuss.

Question 3: [2 marks]

Differentiate between GUI and CLI.

Question 4: [3 marks]

We know that a process generally contains multiple parts, three of which are the Text, the Heap, and the Stack sections. What does each of these sections contain?

Question 5: [4 marks]

Discuss the difference between symmetric and asymmetric multiprocessing.

Question 6: [4 marks]

Answer the following questions:

a) What is the difference between independent and cooperating processes?
b) Mention two reasons why processes might cooperate.

Part 3: Problem Solving: [10 marks]

Question 7: [4 marks]

Suppose that you have a file named **report.txt** that exists in your current directory. Write the commands that accomplish the following tasks:

a) Display the current working directory. (1 mark)
b) Display the following:**-rw--w-r-- 1 admin admin 0 Jan 28 22:39 report.txt** (1 mark)
c) Change the permissions of **report.txt** so that the owner has all permissions, and others have no permissions. (2 marks)

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Question 8: [6 marks]

Write a Bash script that does the following

a) Prompt the user to enter 3 integers (use a loop) (2 marks)

b) Count how many of the entered numbers are negative (2 marks)

c) Compute and displays the sum of the 3 numbers (2 marks)

N.B: Assume all inputs are valid integers (no validation required)

Expected output:
Enter integer 1: 5
Enter integer 2: 3
Enter integer 3: -2
Sum = 6
Count of negative numbers = 1

End of questions

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